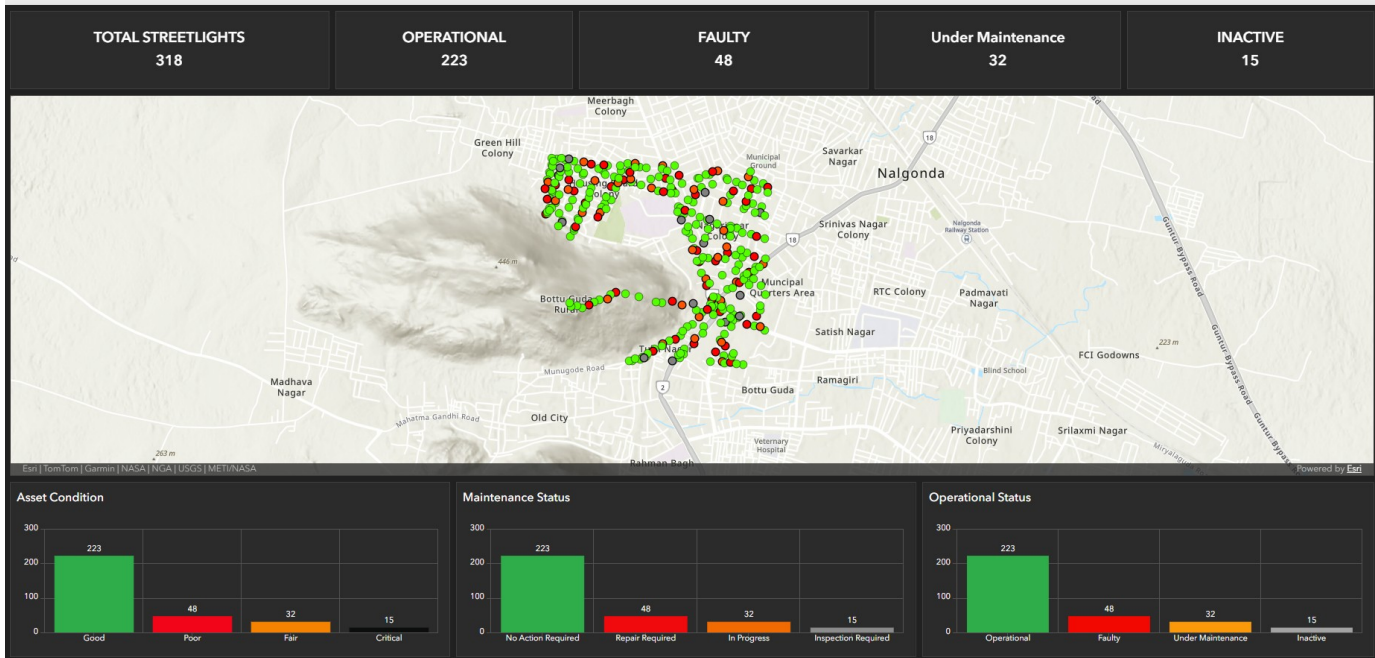


Nalgonda Streetlight Asset Management Dashboard

GIS CASE STUDY | ArcGIS Pro • ArcGIS Online • Web Map • ArcGIS Dashboards

Mohammed Abdul Lateef | Personal GIS Portfolio Project | August 2026



Final dashboard: KPI indicators, interactive streetlight map, and three analytical charts.

Project Overview

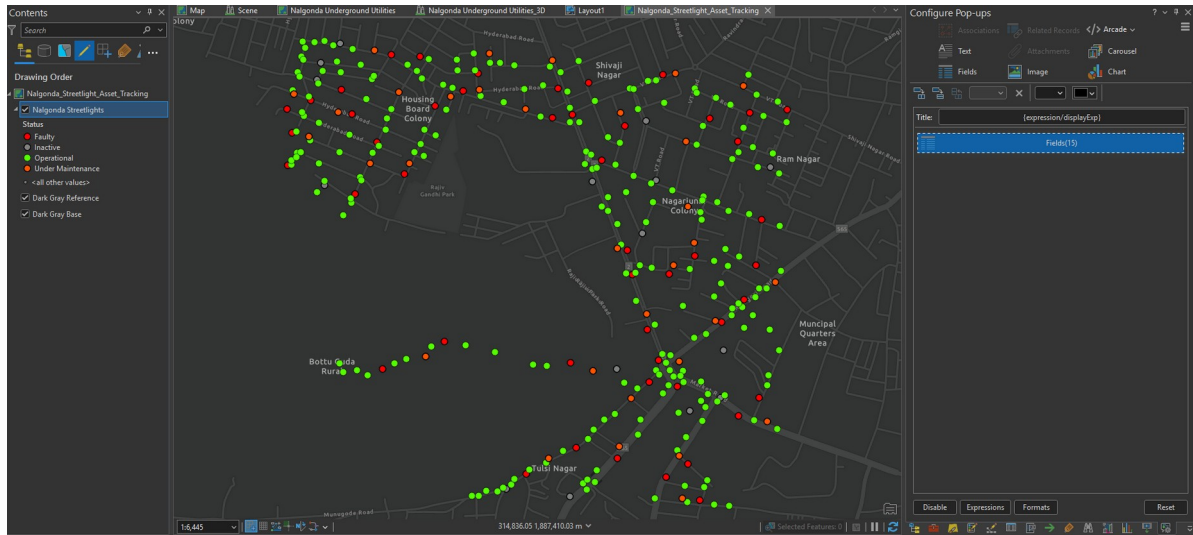
This project demonstrates an end-to-end GIS asset-management workflow for 318 streetlights in Nalgonda. The objective was to transform a point asset layer into a web-enabled monitoring solution that supports operational review, condition assessment, maintenance prioritization, and interactive spatial analysis.

TOTAL	OPERATIONAL	FAULTY	UNDER MAINT.	INACTIVE
318	223	48	32	15

Business / GIS Value

- Provides a single spatial view of streetlight operational status and maintenance needs.
- Links asset condition, maintenance status, and operational status through dashboard cross-filtering.
- Demonstrates a practical Web GIS workflow suitable for infrastructure asset-management portfolios.

1. ArcGIS Pro - Asset Preparation & Smart Pop-ups



ArcGIS Pro map using status-based symbology for the streetlight asset layer.

Data Model and Symbology

The streetlight point layer was prepared in ArcGIS Pro with fields supporting asset identification, road class, pole height, lamp type, installation year, QA status, inspection dates, condition, remarks, and maintenance status. Status-based symbology was applied so operational conditions could be recognized immediately on the map.

- Operational - 223 (green)
- Faulty - 48 (red)
- Under Maintenance - 32 (orange)
- Inactive - 15 (gray)

Professional Pop-up Configuration

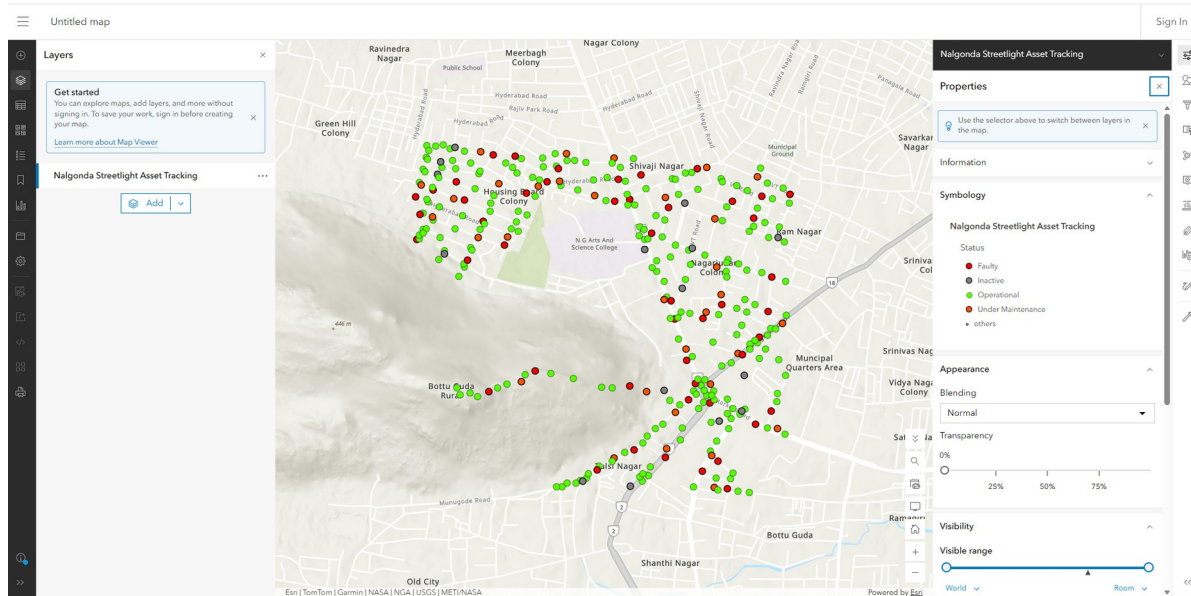
A custom STREETLIGHT ASSET INFORMATION section was configured using dynamic fields for Asset ID, Status, Condition, and Maintenance Status. Unnecessary system and legacy fields were hidden, producing a cleaner asset card for web use. Dynamic-field testing confirmed that selecting a different streetlight refreshed the asset information correctly.

2. Publishing to ArcGIS Online & Web Map

Publishing Workflow

The prepared streetlight layer was shared as a hosted Feature Layer to ArcGIS Online. Before publishing, ArcGIS Pro Analyzer was used to resolve sharing requirements, including unique numeric IDs and the time-zone definition for date fields. The final analyzer result contained zero errors and zero warnings.

- Hosted Feature Layer: Nalgonda Streetlight Asset Tracking
- Project folder: Nalgonda_GeoAI_Smart_City_2030
- Sharing: public for portfolio demonstration
- Time zone: India Standard Time for inspection/update date fields

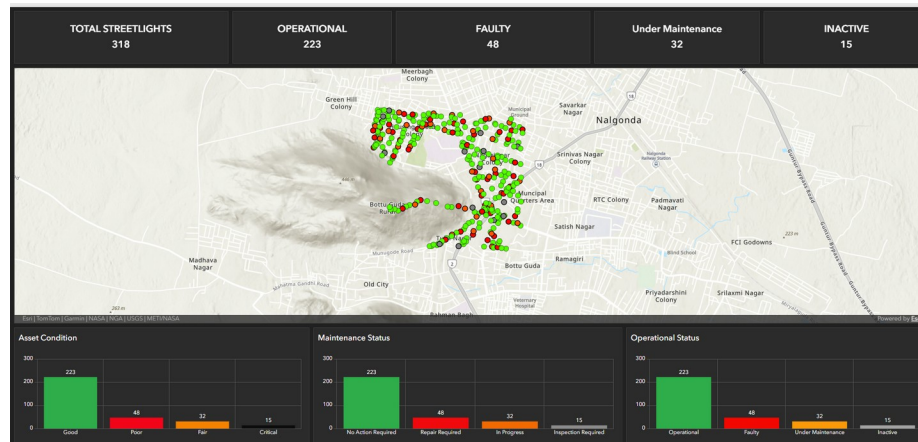


Published layer opened in ArcGIS Online Map Viewer with status symbology retained.

Web Map Validation

The published layer was opened in Map Viewer and validated for geometry, status colors, attribute access, and pop-up behavior. A dedicated web map - Nalgonda Streetlight Asset Tracking - Web Map - was then saved as the data source for the dashboard.

3. Interactive ArcGIS Dashboard & Results

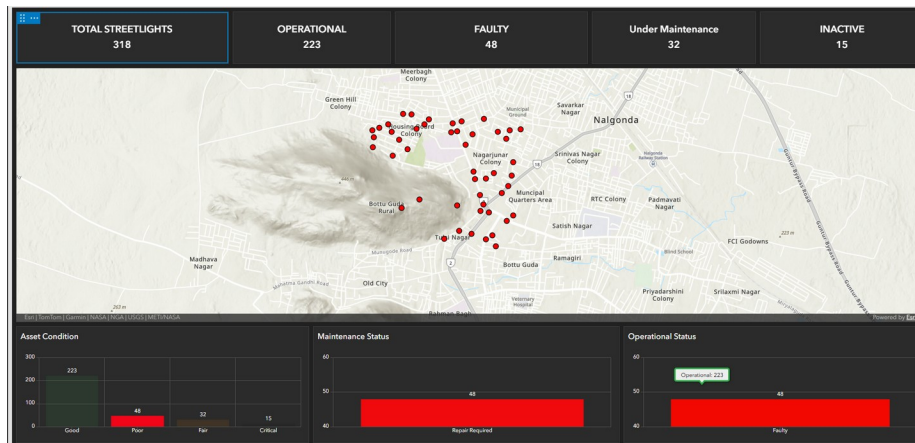


Final dark-theme dashboard with five KPIs and three grouped-value charts.

Dashboard Design

The dashboard was designed with five KPI indicators across the top, a large interactive map, and three grouped-value serial charts: Asset Condition, Maintenance Status, and Operational Status. All values are calculated dynamically from the hosted feature layer rather than manually entered.

- Asset Condition: Good 223 | Poor 48 | Fair 32 | Critical 15
- Maintenance Status: No Action Required 223 | Repair Required 48 | In Progress 32 | Inspection Required 15
- Operational Status: Operational 223 | Faulty 48 | Under Maintenance 32 | Inactive 15



Cross-filter test: selecting Poor (48) filters the map and related charts to the corresponding assets.

Interactive Analysis & Outcome

Dashboard Actions were configured so selections in the analytical charts filter the streetlight map and the related charts. For example, selecting Poor condition isolates 48 assets and simultaneously reveals their linked Repair Required maintenance status and Faulty operational status. This converts the dashboard from a static report into an interactive asset-investigation tool.

Skills Demonstrated

ArcGIS Pro • GIS asset data management • attribute design • categorized symbology • dynamic pop-ups • ArcGIS Online hosted feature layers • Web Maps • ArcGIS Dashboards • KPI indicators • grouped-value charts • dashboard actions • cross-filtering • infrastructure asset monitoring